

1:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
raghu@Raghu:~$ nano practical1.c
raghu@Raghu:~$ gcc practical1.c -o practical1
raghu@Raghu:~$ ./practical1
Enter Linux command: uname
Parent PID : 587
Child PID : 601
Linux
Child process completed.
raghu@Raghu:~$ /practical1
-bash: /practical1: No such file or directory
raghu@Raghu:~$ ./practical1
Enter Linux command: lscpu
Parent PID : 659
Child PID : 660
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13700H
CPU family:                  6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.82
Flags:                        fpu vme de pse tsc msr pae mce
                               cx8 apic sep mtrr pge mca cmov pat pse36 clflush
                               ss ht syscall nx pdpe1gb rdp smp
                               _tsc rep_good nopl xtopology no_stale_pni
                               op_tsc cpuid tsc_known_freq psm
                               sse3 fma cx16 pcid sse4_1 sse4_2
                               opcnt tsc_deadline_timer aes xsave
                               and hypervisor lahf_lm abm 3dnowprefetch
                               brs ibpb stibp ibrs_enhanced
                               d ept_ad fsgsbase tsc_adjust
                               2 erms invpcid rdseed adx smad sha_ni
                               xsaveopt xsavec xgetbv1
```

```

2 erms invpcid rdseed adx smad
sha_ni xsaveopt xsavec xgetbv1
vnmi umip waitpkg gfni vaes v
vdiri movdir64b fsrms md_clear
sh_lld arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 384 KiB (8 instances)
  L1i: 256 KiB (8 instances)
  L2: 10 MiB (8 instances)
  L3: 24 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass;
  ia prctl
  Spectre v1: Mitigation; usercopy/swapgs buffers
  pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic
  dditional; PBRSE-eIBRS SW sequ
  S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
Child process completed.

```

```
raghu@Raghu:~$ ./practical1
```

```
Enter Linux command: ps
```

```
Parent PID : 1067
```

```
Child PID : 1070
```

```
    PID TTY          TIME CMD
    340 pts/0        00:00:00 bash
    1067 pts/0        00:00:00 practical1
    1070 pts/0        00:00:00 ps
```

```
Child process completed.
```

```
raghu@Raghu:~$ ./practical1
```

```
Enter Linux command: top
```

```
Parent PID : 1080
```

```
Child PID : 1081
```

```
top - 09:59:57 up 48 min,  1 user,  load average: 0.01, 0.01, 0.01
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 st
MiB Mem :  7787.6 total,  7107.6 free,   572.6 used,
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	24556	15416	11488	S	0.0	0.0
2	root	20	0	3180	2204	2072	S	0.0	0.0
6	root	20	0	3180	2044	1940	S	0.0	0.0
43	root	19	-1	50396	16712	15428	S	0.0	0.0
78	systemd	20	0	22540	14580	11056	S	0.0	0.0

2.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
raghu@Raghu:~$ nano hello3.c
raghu@Raghu:~$ gcc hello3.c -o hello3
raghu@Raghu:~$ ./hello3
Cannot open source file
raghu@Raghu:~$ nano hello3.c
```

```
raghu@Raghu:~$ nano input.txt
raghu@Raghu:~$ gcc input.c -o input
cc1: fatal error: input.c: No such file or directory
compilation terminated.
raghu@Raghu:~$ nano input.txt
raghu@Raghu:~$ gcc input.txt -o input
/usr/bin/x86_64-linux-gnu-ld.bfd:input.txt: file format not recognized
/usr/bin/x86_64-linux-gnu-ld.bfd:input.txt:0: syntax error, bad operand
collect2: error: ld returned 1 exit status
raghu@Raghu:~$ gcc input.c -o input
cc1: fatal error: input.c: No such file or directory
compilation terminated.
raghu@Raghu:~$ gcc hello3.c -o hello3
raghu@Raghu:~$ nano hello3.c
raghu@Raghu:~$ ./hello3
File copied successfully.
raghu@Raghu:~$ |
```